

Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

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Abstract

A language model is trained to predict text and should be judged by how well it predicts the text it will be used on. Base models are instead chosen by public benchmarks, which vendors optimise for, which can leak into training data, and which score a model before the adaptation every team applies. We return to the model’s own objective. Every candidate is adapted to fresh, decontaminated text from the target domain—its hyperparameters searched per model, then trained to convergence; for compute, the runs reported here use one hand-picked recipe at a fixed budget—and ranked by held-out loss in bits per byte, with no labels. Across seventeen base models and four corpora, adaptation reorders the models, lifting a few that look weak out of the box to the top. The adapted ordering converges on HellaSwag, the likelihood-scored benchmark, turns toward GSM8K and MMLU-Pro when the text is mathematics, and stays stable across seeds, hardware and domains. Ranked before any task exists, fifteen models land within one rank of where their fine-tuned news systems finish (Spearman 0.99); HellaSwag follows at 0.97, while GSM8K and MMLU-Pro manage 0.59 and 0.71. Code and artifacts are released.

1 Introduction

Everything a base model can do, it learned by predicting the next piece of text. The most direct way to ask how good one is, for a given use, is therefore to measure how well it predicts the text of that use. That is not how base models are chosen. A new one ships most weeks, and what arrives with it is a scorecard—MMLU-Pro, HellaSwag, GSM8K—read as the answer to *how good is this model, for what we do?* This paper argues that language models should be evaluated as language models: by the loss they were trained to minimise, on the text they will be used on, after they have been adapted to it.

The scorecard misleads for two reasons. The first is that benchmarks are public. Each was written by other people, for other purposes, and released before the model existed; vendors optimise for the ones they report, and their items and subject matter can reach pre-training data. A high score is therefore consistent with a genuinely stronger model or with one that saw more of that exam, and because pre-training mixtures are not disclosed, no reader can tell which. Inside our own cohort, Llama-3.2-1B and Qwen2.5-1.5B are 2.1 points apart on HellaSwag and 3.4% apart in adapted bits per byte, yet 55.3 points apart on GSM8K: two models that model fresh English almost identically well differ by nearly a factor of ten on one exam. The second reason is timing. Teams rarely deploy a base model exactly as it ships; they adapt it to their data first, so what matters is how good a model *becomes*. A model that has simply never seen prose like the target domain’s can look weak out of the box and still overtake one that looked stronger. Gemma-4-12B is the case in point (Figure 1A): of fifteen models it is ninth on GSM8K, eighth on MMLU-Pro and thirteenth on raw predictive loss, second once adapted, and it builds the best fine-tuned system of all.

There is no universal ground truth for which base model is “better”; a task tells us only which model is better at that task, and tasks disagree. Loss is the principled yardstick. Every base model was trained to minimise cross-entropy on text, so held-out loss is each model’s own objective measured out of sample, and

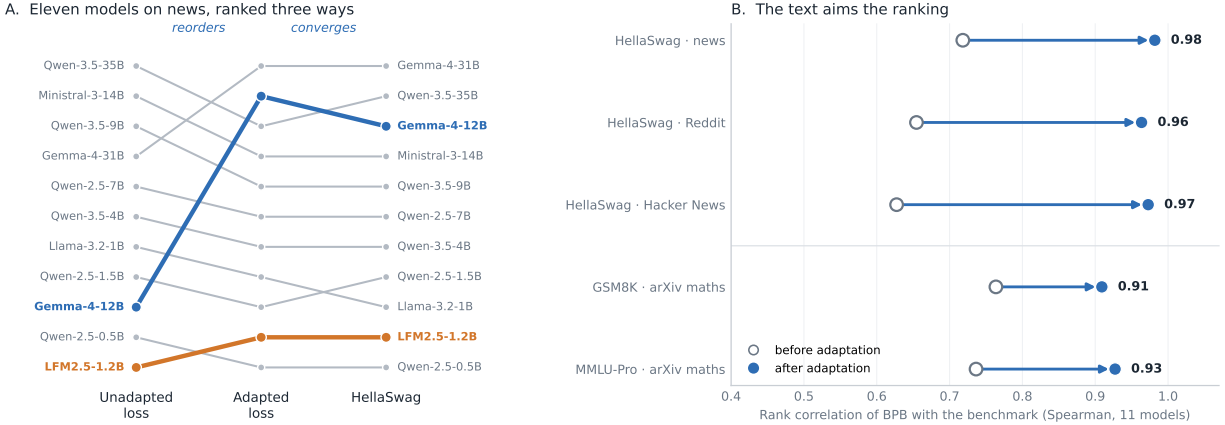


Figure 1: Adaptation reorders the models, and the new order is the language benchmark’s. **A:** the eleven models ranked by unadapted news loss, by adapted news loss and by HellaSwag. Between the first two columns the lines cross: Gemma-4-12B climbs from ninth to second and the Liquid model leaves last place. Between the last two they run almost parallel. **B:** rank correlation between BPB and each benchmark before adaptation (hollow) and after (filled). Adapted on general text, the ordering converges on HellaSwag on all three corpora; adapted on arXiv mathematics, it swings to the reasoning benchmarks.

every other ability it has was learned through that objective. Cross-entropy is also a proper scoring rule: in expectation over text drawn from the domain’s distribution p , a model q scores $H(p, q) = H(p) + \text{KL}(p \| q)$, and $H(p)$ is the same for every model, so the difference between two models’ expected losses is *exactly* the difference in how far each is from the distribution that produced the text; held-out loss estimates it [1]. Taking that seriously asks three things of the measurement. The loss is counted in *bytes*, because every model counts in its own tokens and bytes are the one unit they all share. It is computed on *fresh, decontaminated* text, so that no candidate has trained on it. And it follows *equal adaptation*: every candidate is adapted to the domain by the same procedure—its own hyperparameter search, then training to convergence—so each model is measured as it will be used rather than as it ships. Train-before-test [15] also prepares candidates identically before comparing them, but with labelled benchmark data; we do it unsupervised, on fresh text, and in bytes.

We apply the protocol to seventeen base models from eight families (0.5B–35B) on four corpora, and four things follow. **Adaptation reorders the models.** The unadapted and adapted rankings agree only at Spearman 0.61–0.68 on general text, because unadapted loss mixes familiarity with a style of prose into capability; the only models that climb are the three adaptation helps most (Section 3.1). **The adapted ordering converges on the language benchmark, and can be aimed.** It matches HellaSwag at Spearman 0.96–0.98 on news, Reddit and Hacker News, up from 0.63–0.72, and adapted on mathematics it swings to GSM8K and MMLU-Pro (0.91 and 0.93; Figure 1B, Section 3.2). **It is stable where benchmarks are not.** Seed-to-seed variation is a thousandth of the spread between models, cells reproduce across hardware, and the ordering carries across domains, while identical weights move GSM8K by up to 0.69 points when only the host changes (Section 3.3). **Minutes of adaptation anticipate hours of fine-tuning.** A team will fine-tune its chosen model anyway; measured before any task exists, adapted bits per byte ranks fifteen models at Spearman 0.99 with where their fine-tuned systems finish—every model within one rank, eleven exactly. HellaSwag, the likelihood benchmark, follows at 0.97; the reasoning benchmarks manage 0.59 (GSM8K) and 0.71 (MMLU-Pro) (Section 3.4). Section 2 gives the protocol and Section 3 the results in this order. Every number recomputes from a committed artifact: <https://github.com/DavidAfonsoValente/entropy-bench>.

2 The Protocol

The measurement runs in three phases, and the order is load-bearing: the corpus is fixed and screened before any model touches it.

Phase 1: a decontaminated corpus. Evaluation text must be text no candidate has seen, and we guarantee it three ways. *Time*: the text is dated after every candidate’s release—Reddit and Hacker News are scraped in August 2026, after every model in the cohort shipped, and the arXiv titles and abstracts run from 9 June to 20 August. *Hygiene*: exact duplicates, text failing the Gopher quality heuristics and MinHash/LSH near-duplicates at Jaccard 0.60 are removed, and the 80/10/10 split shares no identifier or normalised text across splits, so nothing in the test split reappears in training. *Membership screens*: the news snapshot (Google News, 8 June 2026, 119,054 articles) predates part of the release window, so every model is also tested for having seen it. Three reference-free tests look for the signatures memorised text leaves: Min-K++ [10] flags passages whose least likely tokens are unusually likely for the model; context sensitivity flags passages whose loss falls more than expected when the model is shown other text from the same corpus; and a perturbation test checks whether the model prefers the original passage over copies with swapped words. The *union* of everything any model flags—80 sequences—is removed for every model, so every candidate is measured on one identical corpus (Appendix C).

Phase 2: equal adaptation. Every candidate is adapted to the retained training text, packed into contiguous 512-token blocks, with LoRA [25] on all linear layers, and every candidate is adapted by the same procedure. Its hyperparameters come from its own Optuna search [24]: a TPE sampler proposes the learning rate, batch size, warm-up, weight decay, LoRA rank, α and dropout; successive halving prunes weak trials from 50 steps on; and every trial is scored by BPB on the validation split, never the test split. Every model receives the same number of trials and no time limit: a search capped by wall-clock time lets small models finish more trials than large ones and so tunes them better, a size bias we observed directly (Appendix C). The chosen configuration is then trained until validation BPB stops improving, so each candidate is measured near the best it can reach on the domain. The released pipeline implements this procedure (`-hparams sweep -no-time-limit`), and also accepts a hand-picked configuration in place of the search.

Searching seventeen models of up to 35B parameters in full was beyond our compute budget, so the results reported here use that second path: one hand-picked configuration for every candidate—LoRA rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32—and a fixed 250 update steps rather than training to convergence. That is 4.1M training tokens per candidate, which on one A100 cost 3.7 GPU-minutes at 0.5B, 11.6 at 7.7B, 24.7 at 12B and 62.7 at 31B. The configuration sits where our own earlier search settled—rank 16 won for eight of the eleven models and the median winning learning rate was 1.1×10^{-4} —and its endpoint is stable to seed and hardware (Section 3.3), but it suits no architecture perfectly, and the full procedure can only tune each candidate closer to its best.

Phase 3: bits per byte. Cross-entropy on the held-out test split is divided by the UTF-8 byte length $|B|$ of the text those predictions cover:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln 2} \times \frac{|\text{Tokens}|}{|B|} \quad (1)$$

The byte count is a property of the text and identical for every candidate, so a larger vocabulary earns no advantage the way it would under per-token perplexity. Test blocks are independent and non-overlapping, so context resets at every cut, and injected document-start markers stay available as context but are always masked as prediction targets. We report both endpoints, before adaptation and after, because the difference between them is itself informative (Section 3.1).

Model	Params (B)	News BPB (rank)		Moves	Gain	Benchmark score, % (rank)		
		unadapted	adapted			HellaSwag	GSM8K	MMLU-Pro
Gemma-4-31B	31.4	0.752 (4)	0.624 (1)	↑3	17.0%	87.9 (1)	85.0 (3)	57.1 (2)
Gemma-4-12B	12.0	0.866 (9)	0.653 (2)	↑7	24.5%	84.0 (3)	64.8 (7)	44.8 (7)
Qwen-3.5-35B-MoE*	34.7	0.707 (1)	0.678 (3)	↓2	4.1%	85.6 (2)	86.6 (1)	61.4 (1)
Ministral-3-14B	14.0	0.718 (2)	0.679 (4)	↓2	5.4%	82.9 (4)	80.8 (6)	52.8 (4)
Qwen-3.5-9B	9.0	0.740 (3)	0.708 (5)	↓2	4.4%	81.6 (5)	85.8 (2)	55.6 (3)
Qwen-2.5-7B	7.7	0.758 (5)	0.717 (6)	↓1	5.4%	80.3 (6)	81.3 (4)	47.2 (6)
Qwen-3.5-4B	4.2	0.782 (6)	0.749 (7)	↓1	4.2%	77.2 (7)	81.0 (5)	52.4 (5)
Llama-3.2-1B	1.2	0.811 (7)	0.771 (8)	↓1	4.9%	65.8 (9)	6.4 (11)	11.2 (11)
Qwen-2.5-1.5B	1.6	0.833 (8)	0.797 (9)	↓1	4.3%	68.0 (8)	61.6 (8)	25.6 (8)
LiquidAI-LFM2.5	1.2	1.340 (11)	0.852 (10)	↑1	36.4%	60.9 (10)	54.1 (9)	16.4 (9)
Qwen-2.5-0.5B	0.5	0.935 (10)	0.901 (11)	↓1	3.6%	51.8 (11)	34.3 (10)	16.0 (10)

Table 1: The eleven models, ranked every way. Ordered by adapted news BPB (lower is better). *Moves* is how many places adaptation shifts a model; *Gain* is the loss reduction it buys. Three models gain far more than the rest—the two Gemmas and the Liquid model—and they are the only three that climb. Once adapted, the order is HellaSwag’s almost exactly, while GSM8K and MMLU-Pro scatter: Llama-3.2-1B, eighth by BPB and ninth on HellaSwag, is last on both. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters.

Cohort and setup. We evaluate 11 base checkpoints spanning dense Transformers, one sparse MoE and one Liquid architecture (0.5B–35B parameters; Table 8) on all four corpora, and extend the news cohort to 17 on preregistered size-adjacency and publisher-spread criteria (Table 7), from the Llama, Qwen, Gemma, Ministral, LiquidAI, Falcon, Mistral and SmolLM families [26, 27, 28, 29, 30]. For comparison every model is also scored on MMLU-Pro [18], HellaSwag [19] and GSM8K [20] with *lm-evaluation-harness* 0.4.12 [12] on one host and one build, under the protocol of Appendix F—directly observed scores rather than vendor-reported card values, with the harness logs published so each recomputes from raw output. Adaptation ran on Leonardo (CINECA), 3 nodes $\times$ 4 A100 64 GB. Intervals resample models, never pairs, because pairs that share a model are not independent (20,000 draws, percentile, clustered); intervals on one fine-tuned pair resample its 500 evaluation articles.

3 Results

3.1 Adaptation reorders the models

Adapting every model before measuring does not just lower every score; it reorders the models (Figure 1A, Table 1). On news the unadapted and adapted rankings of the eleven models agree only at Spearman 0.655, on Reddit 0.609 and on Hacker News 0.682, and a model can move seven places. The movement is not spread evenly. Adaptation lowers loss by 3.6–5.4% on news for every Qwen, Llama and Ministral model, but by 17.0% for Gemma-4-31B, 24.5% for Gemma-4-12B and 36.4% for LFM2.5-1.2B—and those three are the only models that move up; every other model slips one or two places to make room. Gemma-4-12B is ninth of eleven unadapted and second adapted, a 12B model the unadapted reading puts below two models under 2B; LFM2.5-1.2B, the one non-Transformer, is dead last unadapted at 1.34 bits per byte against a 0.5B Qwen’s 0.93, and ahead of it after adaptation.

The reason is familiarity. Zero-shot loss conflates two things that need separating: *this model already writes like my corpus* and *this model is a capable model*. Figure 2 shows the confound in the raw loss, resolved by position inside a packed evaluation block. Because blocks are independent, every block is a cold start. Unadapted, Gemma-4-12B opens a block at 2.83 bits per byte where Qwen-2.5-1.5B—eight times smaller—opens at 2.04, and it stays behind for roughly the first half of the block; by the end of the block it has passed it, 0.72 against 0.77. A block-averaged number therefore reports mostly how quickly

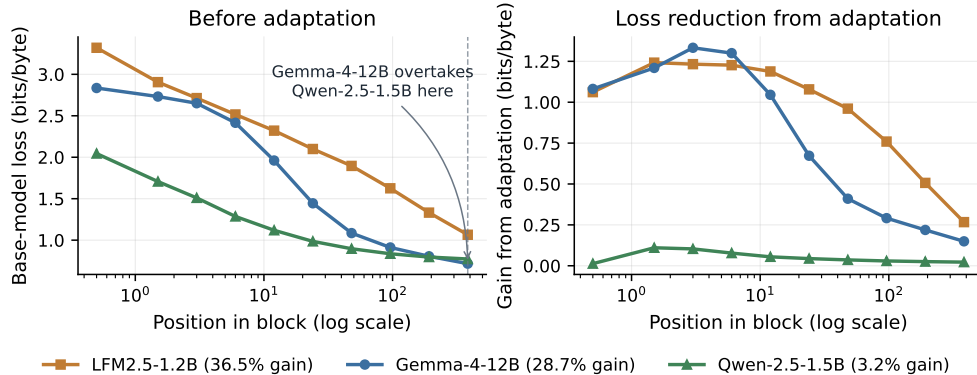


Figure 2: Why the unadapted number mis-ranks: it is largely scoring cold starts. Base loss by position within a packed 512-token evaluation block. Unadapted, Gemma-4-12B opens a block at 2.83 bits per byte against Qwen-2.5-1.5B’s 2.04—a 12B model losing to a 1.5B one—and closes it at 0.72 against 0.77. A block average over-weights the opening, and that is *part* of what puts Gemma-4-12B ninth of eleven on zero-shot BPB. *Right:* the loss reduction adaptation buys at each position is largest in the opening but does not vanish after it—adaptation teaches the corpus’s opening conventions, and does more besides. The right panel uses the three adapters of the earlier per-model search (Appendix C), so its total gains differ from the fixed-recipe reductions quoted in the text; the pattern by position is what it shows.

a model finds its footing in unfamiliar prose—the document conventions, formatting and register of this corpus—which is a real property but not the one a choice of model turns on. A short equal adaptation supplies exactly the missing familiarity, and the reduction it buys is largest in the opening but does not vanish after it: what is left is closer to capability. Unadapted loss is not wrong, then, but it answers a different question—how much this model already resembles the target text—and the models it penalises most are those whose pre-training prose differs most from it, which is also why a non-Transformer with its own training mixture is hit hardest.

3.2 The adapted ordering converges on the language benchmark, and can be aimed

Of the three public benchmarks, HellaSwag is the one closest to language modelling: it asks which ending of a passage is most natural, and the harness scores it by the likelihood of that ending. Before adaptation, BPB on news, Reddit and Hacker News ranks the eleven models at Spearman 0.63–0.72 with HellaSwag; after adaptation the same rank correlation is 0.96–0.98, and on news the adapted ranking orders 53 of HellaSwag’s 55 pairs the same way (Tables 1 and 2). Three independently collected corpora arrive at the same ordering once familiarity is removed, and it is the ordering of the benchmark that measures the same thing. Because HellaSwag and BPB are both likelihoods, this is consistency rather than a second opinion: measuring on fresh text of one’s own gives up nothing the likelihood benchmark captures, while escaping everything that makes a public benchmark suspect.

The effect is specific to what the text is about. On general text the correlation with GSM8K falls on all three corpora, and with MMLU-Pro on two of them; adapted on arXiv mathematics instead, BPB swings toward the reasoning benchmarks—Spearman 0.91 with GSM8K and 0.93 with MMLU-Pro, up from 0.76 and 0.74—and ranking by mathematics BPB picks GSM8K’s own top model at zero regret. The choice of text points the measurement at a capability: the same instrument, fed news, recovers the language benchmark, and fed mathematics, recovers the reasoning ones. Benchmarks cannot be aimed this way, because each is fixed; a corpus is chosen by whoever is choosing the model.

Corpus	Reading	Rank correlation (ρ) with			Pairs ordered as GSM8K
		GSM8K	MMLU-Pro	HellaSwag	
News	zero-shot	0.800	0.845	0.718	45/55
	adapted	0.700	0.773	0.982	42/55
Reddit	zero-shot	0.745	0.782	0.655	43/55
	adapted	0.655	0.745	0.964	41/55
Hacker News	zero-shot	0.800	0.809	0.627	44/55
	adapted	0.745	0.855	0.973	44/55
Math (arXiv)	zero-shot	0.764	0.736	0.482	43/55
	adapted	0.909	0.927	0.818	49/55

Table 2: Adaptation turns the ranking toward the benchmark that matches the text. Rank correlation of BPB with each benchmark over the eleven models, before and after adaptation, on every corpus. On the three general corpora adaptation lifts HellaSwag from 0.63–0.72 to 0.96–0.98 while GSM8K falls; on arXiv mathematics it lifts GSM8K and MMLU-Pro instead. The last column counts the 55 model pairs ordered the same way as GSM8K.

3.3 It is stable where benchmarks are not

A ranking is only as useful as it is repeatable. Three preregistered seed replicates of all 17 news cells put the median within-model standard deviation of adapted BPB at 0.000087 against a between-model spread of 0.074340, a ratio of 0.0012, and not one of the 40 size-matched pairs (models within $2\times$ in parameters) reverses its order across seeds; the one model where seed noise is visible at all is Gemma-4-12B, by up to 1.24%. Re-running two cells weeks later on different hardware reproduced zero-shot BPB exactly and adapted BPB to within 0.0003, and a cell re-run through a different software stack reproduced to 0.01%. The ordering also carries across domains: adapted rankings agree at $\rho = 0.991$ between news and Reddit, 0.955 between news and Hacker News and 0.964 between Reddit and Hacker News (Figure 3). The unadapted ordering is just as stable across domains—0.982, 0.964 and 0.973—so adaptation does not buy stability; it buys a *different* stable ordering, the one Section 3.1 explains and Section 3.2 shows converging on the language benchmark.

Benchmark scores do not have this property. Identical released weights move -0.60 to $+0.69$ GSM8K points when only the host changes, A100 to H100, and one decoding setting a checkpoint ships with was worth 9.86 GSM8K and 14.30 MMLU-Pro points to Gemma-4-12B in our re-score (Appendix F). A generative benchmark measures the model, the harness, the decoding configuration and the hardware at once; bits per byte on a fixed corpus measures the model.

3.4 Minutes of adaptation anticipate hours of fine-tuning

Whichever model a team chooses, it will then train that model for its task. The practical question is whether the cheap measurement already says how that will go. We fine-tuned fifteen of the seventeen models—every one that fits a single GPU—on a news task, writing the opening paragraph of an article from its body, with one identical LoRA recipe and ROUGE-L against the human-written lede on 500 held-out articles from a split the BPB test blocks do not touch; the analysis was fixed in code before the jobs ran (Appendix D). Over all 39 size-matched pairs the fifteen systems form, adapted BPB, measured with no task and no labels, orders 0.949 of the pairs as the fine-tune does, $[0.817, 1.000]$; HellaSwag orders 0.872, while GSM8K and MMLU-Pro sit at 0.474 and 0.513—a coin (Figure 4, Table 3). On the pairs the fine-tune itself separates it agrees in 21 of the 22, and its ranking of the fifteen correlates with their final order at Spearman 0.99, every model within one rank and eleven exactly, where raw loss manages 0.71, GSM8K 0.59, MMLU-Pro 0.71 and HellaSwag 0.97. The lead over ranking by size resolves under this recipe, $+0.231$ $[+0.059, +0.452]$. A

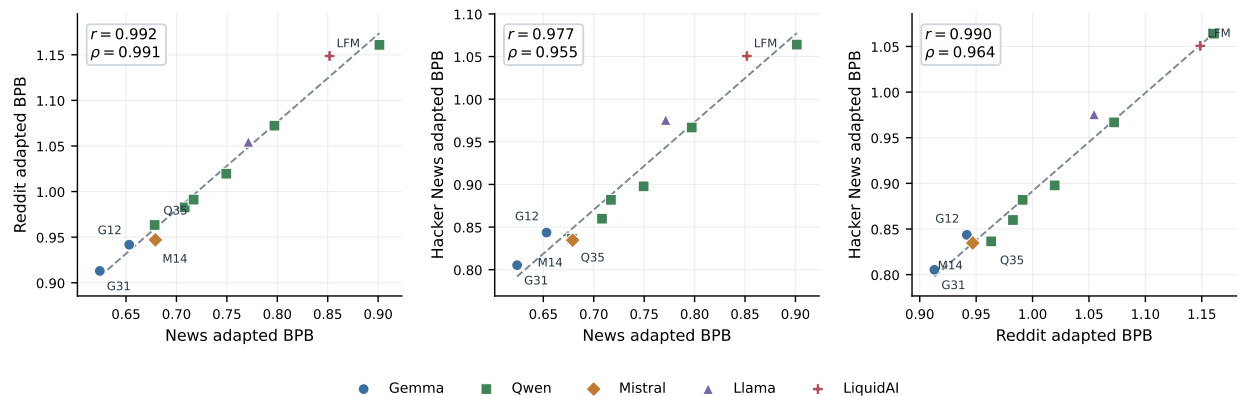


Figure 3: Pairwise adapted-BPB agreement across domains. Absolute difficulty changes, but points remain close to a monotone shared ordering: Pearson correlations between score values are 0.992 (news–Reddit), 0.977 (news–Hacker News) and 0.990 (Reddit–Hacker News). G31/G12 denote Gemma-4-31B/12B; Q35 is Qwen-3.5-35B-MoE; M14 is Ministral-3-14B.

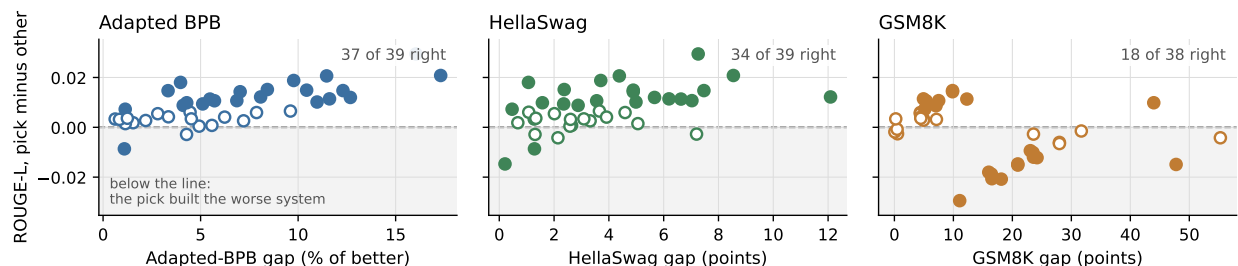


Figure 4: Does each selector agree with the fine-tuned outcome? Every size-matched pair of the fifteen fine-tuned models, oriented so the selector prefers the first: x is how strongly it prefers it, y how much better that model’s fine-tuned system scored. Above the line is a pair the selector orders as the fine-tune does; filled markers are pairs the fine-tune itself resolves. Adapted BPB agrees on 37 of the 39 pairs, and on 21 of the 22 the fine-tune separates; HellaSwag follows at 34. GSM8K agrees on 18 of 38, and every one of its eleven gaps between 15 and 25 points sits *below* the line, on a pair where it favoured the model with the weaker system.

deliberately different recipe—four times the rank, three times the learning rate, half the batch, a new seed—leaves the systems’ ordering almost unchanged (Spearman 0.982) and adapted BPB still first at 0.923, in agreement on all 20 pairs that recipe separates. The two measurements share a domain, so their agreement could be suspected of being news predicting news. It is not: adapted on Reddit instead, BPB orders the nine models that both analyses cover exactly as well as news BPB does (Spearman 0.983 each), Hacker News reaches 0.933, and the same corpora unadapted reach only about 0.6. What carries the prediction is the adaptation, not a shared domain, and that is the point: the few minutes of unlabelled adaptation behind a BPB number already contain most of what hours of labelled fine-tuning reveal, and the benchmarks read before choosing contain almost none of it. **Gemma-4-12B**, 9th and 8th of fifteen on GSM8K and MMLU-Pro, builds the best system of the fifteen, beating Ministral-3-14B by +0.018 [+0.011, +0.025]; and the 55.3-point GSM8K gap between Llama-3.2-1B and Qwen-2.5-1.5B buys no measurable difference once both are fine-tuned. Every model improved on its own base weights and cleared what copying the article’s first sentence scores (0.125).

	All 39 pairs	95% CI	22 resolved pairs	Recipe B
Adapted BPB	0.949	[0.817, 1.000]	0.955	0.923
HellaSwag	0.872	[0.640, 1.000]	0.909	0.846
Zero-shot BPB	0.769	[0.444, 1.000]	0.682	0.744
Parameter count	0.718	[0.463, 0.912]	0.773	0.744
MMLU-Pro	0.513	[0.250, 0.846]	0.500	0.538
GSM8K	0.474	[0.194, 0.824]	0.455	0.447

Table 3: Which selector predicts the fine-tuned system? Fifteen models, each fine-tuned identically on the lede task; the outcome is each system’s ROUGE-L on 500 held-out articles. Each row is a selector’s pairwise accuracy over every size-matched pair, with the model-level cluster bootstrap, and its accuracy on the 22 pairs whose fine-tuned difference is itself resolved by a paired bootstrap over articles. The last column scores the same selectors against systems built with the deliberately different recipe B. GSM8K ties one pair and is scored on 38.

4 Conclusion

Language models should be evaluated as language models: by the loss they were trained to minimise, in bits per byte, on fresh decontaminated text from the domain of use, after the same adaptation for every candidate. There is no ground truth to appeal to, and the case rests on principle—the loss is every model’s own objective, and its differences are differences in distance from the text—and on what the measurement reveals. Unadapted loss confuses familiarity with capability; adaptation removes the familiarity and reorders the models. The adapted ordering converges on the likelihood benchmark on general text (Spearman 0.96–0.98) and on the reasoning benchmarks on mathematics, so it can be aimed at the capability that matters. It is stable to seed, hardware and domain where benchmark scores move with the host. And it anticipates how models fare once fine-tuned, where the reasoning benchmarks do not.

In practice the measurement is cheap enough to run on every candidate. When candidates differ by more than about $2\times$ in parameters the larger one is usually the better choice, and the measurement earns its cost where size has stopped deciding. Between two or three candidates the *gap* is what matters: a different seed moves adapted BPB by up to 1.24% (Gemma-4-12B; the median model, 0.01%) and token-aligned blocks by 0.18%, so we treat a gap under 2% as a tie, to be settled by licence, latency or serving cost. The unadapted number costs one forward pass and is best read as a screen for a candidate badly off-distribution for the target text, not as the choice. Results are comparable only when corpus, split, context, masking and adaptation all match, so the released benchmark keeps one leaderboard per corpus and recipe: a news track run by the maintainers, and a mathematics track on arXiv titles and abstracts whose corpus is public, so that anyone can evaluate a model and submit it.

Three objections deserve answers. *Why not simply fine-tune on the task?* When the task, its labels and the budget exist, that is the better evaluation for that task; a base model is usually chosen before any of them do, and must serve many tasks at once. *Does one hand-picked recipe favour some architectures?* It suits none perfectly, which is why the method searches each model; the reported ordering is stable to seed, and a second fine-tuning recipe leaves the downstream ordering at Spearman 0.982. *Contamination* is addressed by dating, deduplication and membership screens. What remains is breadth: seventeen models is enough to see these patterns and not yet enough to resolve every gap, and a larger, more diverse cohort is the next step.

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A Related Work

From perplexity to comparable loss. Language modelling was grounded in byte- and character-level entropy from the start [1], but token-level perplexity depends on each model’s vocabulary [2, 4]; normalising by UTF-8 bytes restores a common unit [3, 5], and loss-based metrics carry better signal and lower noise than accuracy across 465 models [13]. Our contribution is the operational protocol around the metric, not the metric.

Where the loss is measured matters. Thrush et al. turn the varying correlation between per-domain perplexity and benchmark performance into a pretraining-data selection rule [14]. Their estimand is the complement of ours: they fix the benchmark suite and choose data to *train* one strong model, where we fix the candidate pool and ask how the *selection ranking* moves.

Equal preparation. Dynamic evaluation updates a model on recent test history [21, 22] and domain-adaptive

pretraining improves fit to a target distribution [23]. Closest to our motivation, *train-before-test* gives 61 models identical supervised preparation across 24 tasks and raises cross-benchmark Kendall agreement from 0.52 to 0.76 [15]; they rank post-fine-tuning accuracy on public labelled benchmarks, where we rank held-out fit after *unsupervised* adaptation to a fresh corpus, across tokenizers. The core idea—prepare every candidate identically before comparing them—is theirs; what we add is doing it without labels, on text no vendor has seen, in a unit that compares tokenizers, and checking the result against real fine-tuned systems. Controlled families show a smooth loss-to-error relation [17, 16]; we test it *across* families.

Contamination. Models memorise duplicated pretraining text [6, 7], so deduplication and contamination analysis are necessary [8, 11]; reference-free membership signals are useful screens [9, 10] but prove nothing about a proprietary training set, so we use them only for conservative quarantine.

Corpus	Raw records	Documents	Median words	Text	Contamination control
Google News, 8 Jun 2026	119,054	119,054	362	361 MB	Min-K++ / context / perturbation union
Reddit, 6–10 Aug 2026 [†]	3,054,623	141,527	624	533 MB	Post-release scrape
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB	Post-release scrape
arXiv math. *, 9 Jun–20 Aug 2026 [‡]	3,207	3,207	—	15 MB	Post-release scrape + GSM8K shingle screen

Table 4: Corpora. All four receive identical exact-deduplication, Gopher quality filtering and MinHash/LSH near-duplicate removal at Jaccard 0.60; they differ in *contamination* control, deliberately. The reference-free screen ran on the news corpus, whose 8 June 2026 snapshot predates part of the cohort’s release window; Reddit and Hacker News were scraped after every model shipped, as was most of the arXiv set, and they are protected by construction. Reddit and Hacker News comments are grouped to document length. [†]Four days are available in the stated range. [‡]Used only for the mathematics adaptation.

B The Full Benchmark Alignment

Table 2 (Section 3.2) reports all twenty-four (corpus $\times$ reading $\times$ benchmark) cells over the full 55 pairs; Table 5 prices each reading’s top-1 pick against each benchmark’s own. The rotation (Appendix C) is legible column by column: adapting on general text raises HellaSwag agreement and lowers the GSM8K column on every corpus and MMLU-Pro on two of the three general-text corpora; adapting on mathematics reverses the GSM8K half. The pairs-correct column—GSM8K only—shows how little of this is visible in a rank correlation: across its eight rows it spans 41/55 to 49/55.

C Robustness and Measurement Artefacts

Stability across domains. All eleven models ran on three independently collected domains under the same recipe (Tables 4 and 8). Adaptation genuinely changes the choice—zero-shot and adapted rankings agree only at Spearman $\rho = 0.655$ on news, 0.609 on Reddit and 0.682 on Hacker News, with models moving up to seven places—and the adapted ordering then holds across domains: news and Reddit agree on nine of eleven exact ranks ($\rho = 0.991$, maximum displacement one), news against Hacker News gives 0.955 and Reddit against Hacker News 0.964, and the agreement is not merely ordinal (Figure 3). **So does the zero-shot ordering**—0.982, 0.964 and 0.973 on the same three comparisons. The tempting story is that adaptation buys stability, and it does not; it buys a *different* stable ordering, and Section 3.1 is the evidence that it is the better of the two.

Why the search gives every model the same number of trials. The search of Phase 2 uses a multivariate TPE sampler (seed 42) over learning rate (10^{-6} – 10^{-3} , log), physical batch (2, 4, 8), gradient accumulation (1–16), warm-up ratio (0–0.2), weight decay (10^{-5} – 10^{-1} , log), LoRA rank (8–64), α as a multiple of rank (1, 2, 4) and dropout (0–0.2); a successive-halving pruner starts trials at 50 steps and promotes the best third at each rung; and the objective is BPB on a fixed 256-block subset of the validation split. An earlier run of this search on the eleven-model news cohort was capped by wall-clock time rather than by trial count, and the cap was confounded with model size: the two largest models completed four trials each while the 1B models completed 17 and 18 (results/legacy_sweep/*/trials.csv), and the five models of 9B or more all kept the same third draw of the seeded sampler as their winner, while the smaller

Corpus and reading	Picks	Accuracy given up vs. the benchmark’s own pick (pts)		
		GSM8K	MMLU-Pro	HellaSwag
News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
News adapted	Gemma-4-31B	1.6	4.3	0.0
Reddit zero-shot	Ministral-3-14B	5.8	8.6	5.0
Reddit adapted	Gemma-4-31B	1.6	4.3	0.0
Hacker News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Hacker News adapted	Gemma-4-31B	1.6	4.3	0.0
Math (arXiv) zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4

Table 5: What choosing on BPB costs you on each benchmark. For every corpus and reading, the model the ranking puts first and how many accuracy points that costs against each benchmark’s own top-1 model. Only three distinct picks occur in eight rows, which is itself the finding: the adapted ranking selects Gemma-4-31B on every general-text corpus—the HellaSwag leader outright, at 1.6 GSM8K and 4.3 MMLU-Pro points—while adapting on mathematics recovers the GSM8K and MMLU-Pro leader at zero regret. The corpus, not the metric, decides which column you pay in.

models’ winners came from much later in their searches. A time-capped search hands small models a better-tuned adapter and biases the ranking by parameter count, which is why the released pipeline budgets the search in trials and applies no time cap by default. Apart from the right panel of Figure 2, which uses three of its adapters to show where in a block adaptation gains, none of those adapters produces a number in the paper; the reported results use the hand-picked configuration of Phase 2.

Membership screens. The news corpus additionally passes three reference-free membership screens per model—Min-K++ (normalised likelihood of the lowest-probability $K=20\%$ target tokens), context sensitivity (target NLL with and without sampled same-corpus prefixes), and, for flagged candidates only, preference over two word-swap/placeholder perturbations. The global union quarantined 80 unique sequences, after which every model receives the identical retained corpus. Near-100% preference for the unperturbed original is expected—fluency alone produces it—so we draw no memorisation conclusion from it (results/contamination/).

The cold start, seen a second way. Varying the inference context on 64 fixed held-out target windows from 16 to 2048 tokens reproduces Figure 2’s effect non-positionally: both models improve with context and the larger improves faster, so any protocol that scores short contexts under-rates the models that use long ones. Measured context sensitivity is 0.032 bits per byte per doubling, which is the constant the block-alignment bound below uses.

Token-aligned blocks: a real bias, bounded. Evaluation blocks are 512 *tokens* rather than a fixed byte span, so models with different tokenizers score slightly different amounts of text—5.92–6.24 MB across the eleven-model cohort, a 5% spread, and 9.9% across the seventeen. Bounding the resulting bias with the measured context sensitivity puts it at $0.08\times$ the smallest adjacent rank gap in the eleven-model cohort. In the extended cohort the bound is looser and worth stating: on the worst adjacent pair, Qwen2.5-3B over Falcon3-7B, whose BPB gap is only 0.29%, the bound reaches $0.61\times$ the gap. It still **overturms no rank**, but byte-aligned blocking would remove the question, and we recommend it for future work.

Seed variation is negligible. Every adaptation cell ran with the torch RNG unseeded, so a preregistered check re-ran all 17 news models three times (51 cells, everything else pinned). Median within-model SD of adapted BPB is 0.000087 against a between-model SD of 0.074340, a ratio of 0.0012, and *no* pair of models within $2\times$ in size—0 of 40—reverses its ordering across seeds. The spread is not uniform: Gemma-4-12B is the one model where seed noise is visible at all (SD 0.00315, $5.3\times$ the next-largest spread).

Steering: direction, not magnitude. Adapting on general text raises agreement with HellaSwag on news from $\rho = 0.718$ to 0.982 while GSM8K falls from 0.800 to 0.700 and MMLU-Pro from 0.845 to 0.773; adapting on

Model	Out of the box			After	Adapted BPB
	raw loss	GSM8K	MMLU-Pro	<i>fine-tuning</i>	<i>predicted</i>
Gemma-4-12B	13	9	8	1	2
Llama-3.2-1B	10	15	15	11	11
LFM2.5-1.2B	15	12	13	14	14

Table 6: How a model looks is not how it finishes. Rank among the fifteen fine-tuned models (1 is best). The benchmarks misjudge where these models end up, and raw loss misjudges Gemma-4-12B by twelve places: it is mid-table on both benchmarks and near the bottom on raw loss, then builds the best system of all. Llama-3.2-1B is last on both benchmarks and finishes eleventh. Adapted BPB, measured before any task fine-tuning, places every model within a single rank of where it finishes—all fifteen, and eleven exactly.

contamination-screened arXiv mathematics instead raises GSM8K agreement to 0.909 and restores the GSM8K oracle pick at zero regret, while three general-text corpora tried as controls all leave GSM8K at or below the do-nothing baseline. That is the rotation, and it is why the body reports only its direction. Because the steering hypothesis predicts a *joint* movement, the better-powered statistic is a difference in differences—(GSM8K minus HellaSwag selection accuracy under mathematics adaptation) minus the same quantity under a general-text corpus—which is +0.242 pooled. Resampling models gives a strictly positive difference in 93.7% of draws; the stricter publisher-family clustering the mechanism itself implies gives 80.9% with not one negative draw in 20,000, but no longer clears 0.05 on any tie convention. A leave-one-family-out jackknife shows why: dropping the two Gemma models takes the pooled difference from +0.242 to +0.056. The remedy is more *families*, not more models.

Three reproduction controls. Re-running two Ministral-3-14B cells weeks later on different hardware reproduced zero-shot BPB exactly and adapted BPB to within 0.0003, two orders of magnitude below the differences compared. Because the extension models were adapted on a different software stack, a published cell (Qwen2.5-7B) was re-run through the extension pipeline; the analysis refuses to pool the cohorts unless it reproduces within 1%, and it reproduces to 0.01% on adapted BPB (0.7169 against 0.7170). And because benchmark rankings are themselves estimates, recomputing selection accuracy while discarding pairs whose benchmark gap is smaller than k combined standard errors leaves the result intact and slightly improves it: on GSM8K 0.909 at $k=0$ becomes 0.933 at $k=3$, and on HellaSwag 0.836 becomes 0.863, so the metric’s errors concentrate in pairs the benchmark itself cannot separate.

D The Downstream Fine-Tune

Section 3.4 reports the result; this is how it was built, and what it does and does not license. It ran in two stages, each preregistered in code before its job returned: first the two pairs on which the selectors disagree most (Gemma-4-12B against Ministral-3-14B, Llama-3.2-1B against Qwen-2.5-1.5B), then every cohort model that fits one GPU, so that no pair is chosen by anyone. Gemma-4-31B and Qwen-3.5-35B-MoE need more than one GPU *for this fine-tune*—its 1,280-token sequences and its generation pass cost far more memory than the 512-token adaptation priced in Section 1—and are excluded by that rule, which costs one in-band pair of forty.

The task. Journalistic convention makes a news article’s first paragraph a standalone summary of it, so the corpus supplies its own supervision: the input is the article with its opening paragraph removed, truncated to 3,000 characters, and the target is that paragraph. Articles qualify if they have at least three paragraphs, an opening paragraph of 200–700 characters ending in sentence punctuation, and at least 800 characters of remaining body. Of the 11,905 validation documents, 3,627 qualify; 2,000 are sampled for training and 500 for evaluation under a fixed seed.

Why the validation split. The BPB test blocks live in the *test* split, so the fine-tuning task is built from the validation split instead. That is verified rather than assumed: the test split reproduces from an independent reconstruction, and the adaptation pipeline’s contamination pass—which would have split the corpus differently—is off by default and recorded as off in every published cell. Evaluation documents are then hashed against the test documents twice, on full normalised text and on a 400-character opening, the second because a syndicated wire story is republished with an edited body and an identical lede. Four articles were dropped by that second check and none by the first.

Equal preparation. Both members of a pair get the paper’s own adaptation recipe unchanged —LoRA rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, 250 steps—with prompt tokens masked out of the loss so the model is trained to write the lede rather than to reconstruct the article. Nothing is tuned on either model, because a per-model search is exactly the cost this protocol exists to avoid. The tokenizers differ, so the budget is matched in update steps $\times$ batch $\times$ sequence length; each model’s realised token count is in the artifact. Decoding is greedy with a 180-token cap and each checkpoint’s own generation configuration otherwise untouched, following Appendix F. The prompt is assembled so that the instruction always survives truncation: the *body* is cut to fit, never the instruction after it, because right-truncating the assembled prompt would delete the instruction from the longest articles at a different article count for each tokenizer—an asymmetry between the two models being compared. At most four of the 500 evaluation articles were cut for any model, and none for three of them.

The decision rules, fixed before each run. For the two chosen pairs: mean ROUGE-L F1 over the 500 held-out articles, a paired bootstrap over articles (20,000 draws, 95% percentile interval) and a practical-equivalence margin of 0.01—an interval excluding zero resolves the pair for whichever reading predicted that winner. The first pair resolved for adapted BPB; the second was inconclusive. For the full cohort: the fine-tuned ROUGE-L is the outcome and every selector is scored with the paper’s own statistic—pairwise accuracy within $2\times$, model-level cluster bootstrap, 20,000 draws (Table 3). A separate guard voids any comparison in which fine-tuning failed to improve a model over its own base weights, since ordering unmoved models would order their base behaviour under another name. It did not fire: all fifteen models gained between +0.041 and +0.153.

A second recipe. Both readers of an earlier draft raised the same objection: the fine-tune reused the selector’s own recipe, so a model that adapts quickly under it could win both. Recipe B changes every setting the two shared—LoRA rank 64, $\alpha=128$, learning rate 3×10^{-4} , effective batch 16, 400 steps, a new seed—on the same task, data, decoding and metric, preregistered before it ran. It fits the training articles far harder and scores slightly below recipe A for every model, yet orders the fifteen systems almost identically (Spearman 0.982). Against recipe-B systems adapted BPB remains the top selector at 0.923 [0.750, 1.000], ahead of HellaSwag at 0.846, parameter count at 0.744 and GSM8K and MMLU-Pro near chance, and it picks the better system on all 20 pairs recipe B resolves. Its margin over parameter count, resolved under recipe A, does not resolve here (+0.179 [−0.036, +0.441]), so that margin should be read as recipe-specific (Table 3).

What copying buys. A news body restates its own lede, so reference overlap is partly available by retrieval. Two reference points bound it: the article’s first body sentence scores 0.125 against the reference and a random body sentence 0.111. Unadapted base weights land close to copying—inspection confirms they largely do copy—while every fine-tuned system clears it, from 0.188 (Qwen-2.5-0.5B) to 0.249 (Gemma-4-12B).

How close the corpus has to be. Exploratory, run after the result at a reader’s request: scoring the news fine-tune with adapted BPB from each of the other corpora, on the ten size-matched pairs of the nine fine-tuned models that have all four, gives 0.900 for news and for Reddit, 0.700 for Hacker News and 0.600 for arXiv mathematics; the unadapted readings reach 0.500–0.600. Fresh general text does most of the work and nearby text does it best. Ten pairs support no interval, so this is a direction, not a result.

What this does not show. One task, one domain, two recipes, one seed each. On this task adapted BPB ordered the fine-tuned systems better than any other selector and GSM8K and MMLU-Pro ordered them at chance; its margin over HellaSwag does not resolve. It is not a claim that adapted BPB predicts downstream utility in general, and the paper does not make one.

E The Cohorts

Table 7 is the extended 17-model cohort behind the seed replicates and, less the two models too large for one GPU, the fine-tune: every model, its parameter count and its adapted news BPB. Table 8 then gives the original eleven across all three general corpora and both BPB endpoints, which is the population every other analysis in the paper uses. Table 8 is the full three-corpus leaderboard: both endpoints for all eleven models on news, Reddit and Hacker News, ordered

Model	Params (B)	Adapted BPB	Cohort
Gemma-4-31B	31.4	0.624	original
Mistral-Nemo-12B	12.3	0.646	added
Gemma-4-12B	12.0	0.653	original
Qwen-2.5-14B	14.8	0.675	added
Qwen-3.5-35B-MoE*	34.7	0.678	original
Ministral-3-14B	14.0	0.679	original
Qwen-3.5-9B	9.0	0.708	original
Qwen-2.5-7B	7.7	0.717	original
Falcon3-10B	10.4	0.728	added
Qwen-3.5-4B	4.2	0.749	original
Qwen-2.5-3B	3.1	0.756	added
Falcon3-7B	7.5	0.758	added
Llama-3.2-1B	1.2	0.771	original
SmolLM2-1.7B	1.7	0.788	added
Qwen-2.5-1.5B	1.6	0.797	original
LiquidAI-LFM2.5	1.2	0.852	original
Qwen-2.5-0.5B	0.5	0.901	original

Table 7: The extended cohort, in full. All 17 models ordered by adapted news BPB. “added” marks the six models admitted for the size-matched extension on preregistered size-adjacency and publisher-spread criteria; the other eleven carry the four-corpus analyses. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters; parameter count here is always the total.

by adapted news BPB. It is the artifact every eleven-model number in the paper derives from, as Table 7 is for the seventeen.

Three checkpoints are excluded. The two OLMo-2 models fell under the preregistered infrastructure-fault rule: their adapted BPB cells exist and are unused, because the generation engine the study’s evaluations share would not initialise for them, identically before and after a cache rebuild. The third, Qwen-2.5-32B, did not finish its 250 steps inside the run window; it would have added about two size-matched pairs at the top of the size range. The OLMo exclusion removed a whole publisher, which is what the cohort most needs.

F The Benchmark Column Re-Score

Harness configuration. MMLU-Pro is 5-shot chain-of-thought with exact match on one stratified 1,000-question subset (seed 42); HellaSwag is 10-shot length-normalised multiple-choice accuracy on all 10,042 validation examples; GSM8K is 5-shot chain-of-thought with flexible numeric extraction on all 1,319 test examples. All bf16, greedy, tokenizer-native BOS.

July’s eleven-model column was scored on a host that no longer exists and generative scores do not pool across hosts—identical released weights move -0.60 to $+0.69$ GSM8K points when only the host changes, A100 to H100—so it was re-scored in full—seventeen models $\times$ three benchmarks = fifty-one cells, one host (A100-SXM-64GB), one build—with the design and the exclusion rule fixed in advance. Task definitions match exactly (17/17 identical task hashes on both hashed benchmarks) and every overlapping cell agrees with July to within 1.3 points, most to within 0.3.

Reaching that took one correction worth reporting, because the fault was in our own fix. Because the newer backend honours each checkpoint’s shipped `generation_config.json` and the original did not, we replaced every checkpoint’s generation settings with one identical policy—which also discarded `suppress_tokens`, a setting that is live for gemma-4-12B. That single omission cost it 9.86 GSM8K and 14.30 MMLU-Pro points while leaving HellaSwag, scored by likelihood and never calling `generate()`, intact. Keeping the setting for all seventeen models reproduces July to $+1.21$ points on GSM8K, $+0.10$ on MMLU-Pro and -0.27 on HellaSwag. That GSM8K cell is the one outside the preregistered 1.0-point tolerance, within the allowance of one that the re-score declared in advance: the column is within its stated allowance rather than clean, and those are different claims. The lesson generalises: a

Model	News		Reddit		Hacker News	
	zero-shot	adapted	zero-shot	adapted	zero-shot	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LiquidAI-LFM2.5	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 8: Marker-masked BPB under one fixed adaptation protocol. Lower is better. Rows are ordered by adapted news BPB; all eleven models occur in all three corpora.

wrapper imposed to remove a confound can be the confound, so the unmodified harness belongs in the design as a control.